Credibility of Synthesized Speech
A Project in Sociolinguistics

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1 Introduction

Media landscapes are undergoing rapid transformation with the rise of Artificial Intelligence (AI). AI-generated news anchors—such as Xinhua's "Xin Xiaomeng" in China and India's "Lisa" and "Sana"—are increasingly being deployed to read and present news content. While these systems offer advantages in efficiency, scalability, and consistency, they frequently lack qualities that characterize human news delivery: empathy, spontaneity, emotional nuance, prosodic variation, and natural rhythm. These limitations raise important sociolinguistic questions about how audiences perceive such voices and whether AI-generated speech can convey authenticity and credibility comparable to that of human anchors.

This motivates the core research question of this study: *Do perceptible linguistic and prosodic differences emerge between human and AI news anchors, and how do listeners interpret these differences in terms of naturalness, credibility, and trust?* By examining listener judgments across multiple languages and demographic groups, we explore whether prosodic cues remain central to distinguishing human speech from synthetic news delivery.

2 Aim

The primary aim of this study is to systematically compare human and AI news anchor speech using perception-based evaluations. We investigate whether prosodic and linguistic features—such as tone, intonation, inflection, stress, and rate of speech—influence how listeners assess the naturalness, credibility, and authenticity of news delivery. Through cross-linguistic and demographic analysis, the study seeks to understand both the strengths and limitations of contemporary AI-generated news speech.

3 Objectives

1. To analyse prosodic differences between human and AI-generated news speech across multiple linguistic dimensions.

2. To examine how variation in prosodic features influences perceived credibility, trustworthiness, and relatability of news delivery.

3. To identify the prosodic attributes that most strongly differentiate AI speech from human speech in listener judgments.

4 Literature Review

Prior studies suggest that prosody plays a key role in newscaster perception. Prosodic cues such as pitch, intonation breaks, speech rhythm, and voice quality help differentiate professional news speech from everyday speech. AI voices struggle with emotional expression and tend to lack non-modal phonation.

Listeners also identify clear boundaries between human and AI voices; while human–human and AI–AI voice matching achieves high accuracy, matching an AI clone to its original human counterpart remains difficult. When audiences are informed that AI is involved in news production, their trust in content credibility decreases.

Furthermore, psychophysiological studies show greater brain activity when listening to human newscasters, suggesting higher engagement. Although some findings indicate comparable levels of credibility between AI and human speech, students and listeners generally find human voices more appealing, with a mild preference for female AI voices.

Several overview studies confirm that while AI can emulate certain linguistic traits and maintain consistent delivery, it struggles to replicate the emotional connection of human hosts. The need for human anchors persists.

5 Methodology

Our methodology consisted of three primary stages: (i) identifying salient linguistic features that differentiate human and AI news speech, (ii) constructing a multilingual dataset of news clips, and (iii) designing and administering a perception survey to evaluate listener judgments.

5.1 Preliminary Analysis and Feature Identification

We began by qualitatively analysing a collection of news clips featuring both human and AI newscasters. Through careful listening, we identified linguistic and prosodic cues that appeared noticeably different in synthesized speech. These included irregularities in intonation patterns, lack of natural variation in tone, mechanical stress placement, and inconsistencies in speech rhythm. These observations motivated the choice of prosodic features incorporated into the survey.

5.2 Data Collection

To build our stimulus set, we sourced audio clips from publicly available YouTube broadcasts of news bulletins. The dataset consisted of both human anchors from various Indian and international news channels and AI-generated anchors. Clips were selected to ensure comparable content, length, and audio quality wherever possible. A balanced dataset was maintained with an enforced 50% human and 50% AI distribution. For certain AI clips, additional sped-up versions were generated to test whether increased speech rate improved perceived naturalness.

In order to minimise external bias in the assessment of credibility and trustworthiness, we exercised care in selecting the content of the news clips. Clips discussing politically charged, sensational, or emotionally loaded topics were avoided, as such content may predispose listeners to form opinions based on topic familiarity rather than linguistic cues. Instead, we selected neutral and non-controversial news segments (e.g., general informational bulletins) so that listeners' judgments would be driven primarily by prosodic and linguistic features rather than the subject matter being reported.

5.3 AI Anchors Used

Prosodic manipulation was tested using original and sped-up forms of AI-generated speech. The synthesized speech stimuli were produced using the following AI-anchor voice models:

- Sana
- Ivan
- Anjana 2.0
- Aashi
- Priya
- Elihay Vidal
- Ariela Karnel
- Meera Rao

These models were selected to represent variation in gender, accent, speaking style, and prosodic quality, enabling a broader examination of synthesized news speech across multiple voices.

5.4 Survey Design

A perception experiment was designed to examine how listeners classify and evaluate human versus AI newscaster speech. The survey was hosted on a custom Streamlit web application to allow for seamless audio playback and structured rating input.

Each participant listened to:

- Four English clips, and
- If linguistically proficient, two additional clips in one of Hindi, Tamil, or Malayalam.

After listening to each clip, participants completed two tasks:

1. **First Impressions:** Rate the trustworthiness,
2. **Prosodic Evaluation Task:** Rate the clip on a 5-point Likert scale for the following linguistic features:
    - Rate of Speech
    - Tone
    - Inflection
    - Stress
    - Intonation

These features were selected based on our preliminary analysis of prosodic differences observed between human and synthesized speech.

5.5 Tools and Implementation

The survey interface was implemented using Streamlit, which allowed for controlled presentation of audio stimuli and structured collection of participant responses. All subsequent data cleaning, aggregation, and visualisation were performed using Python libraries such as `pandas`, `numpy`, and `matplotlib`.

5.6 Participants

80 respondents participated, representing multiple linguistic backgrounds: 36% Hindi, 22.5% Tamil, 18.75% Malayalam. Participants evaluated 4–6 clips depending on language competence. Stimuli included original and sped-up AI voice versions (15–20%).

6 Observations and Patterns

Across participant responses, several consistent patterns emerged regarding the perception of human and AI newscaster speech. These patterns reflect both prosodic differences between the two categories of speech and sociolinguistic variation among listeners.

6.1 Prosodic Feature Ratings

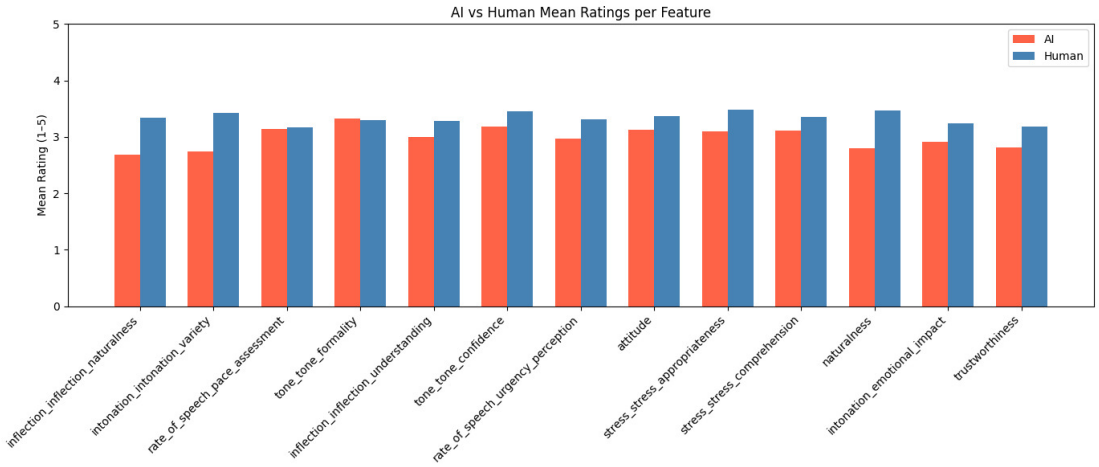

Figure 2: Mean ratings for AI vs. Human speech across prosodic and perceptual features.

Human clips received consistently higher ratings across all prosodic dimensions, including tone, inflection, stress, intonation, and rate of speech. As shown in the figure, participants attributed greater naturalness and expressiveness to human voices, while AI-generated speech was frequently described as monotonic or lacking variation. Tone emerged as the most salient cue, with listeners relying heavily on tonal consistency and expressiveness to judge credibility.

6.2 Feature Importance for Identifying AI Speech

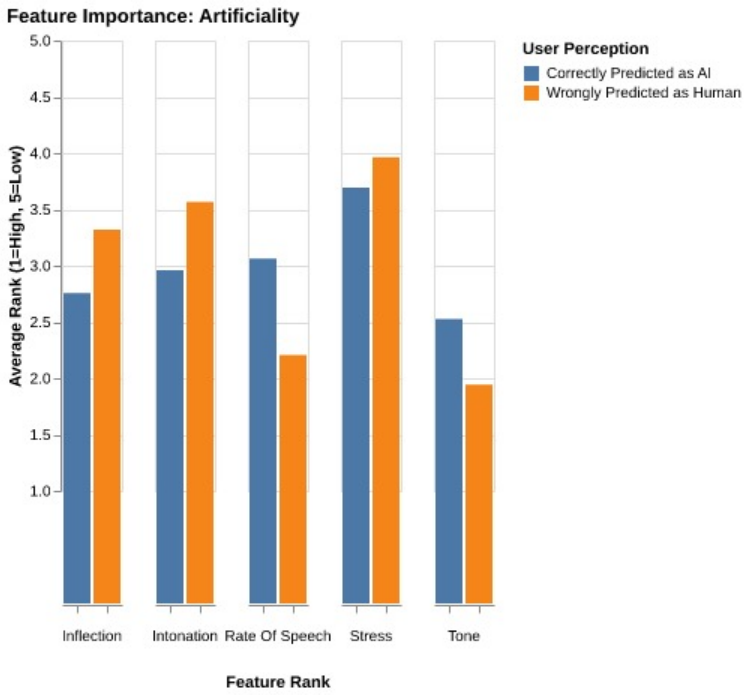

Figure 3: Feature importance comparing correctly and incorrectly identified AI clips.

Analysis of correctly and incorrectly classified clips revealed that inflection and intonation were the most influential predictors when distinguishing AI from human speech. AI voices that mimicked natural inflection patterns were more likely to be misidentified as human, whereas clips with flat or uniform pitch contours were rapidly detected as synthetic.

6.3 Trustworthiness and Prosody

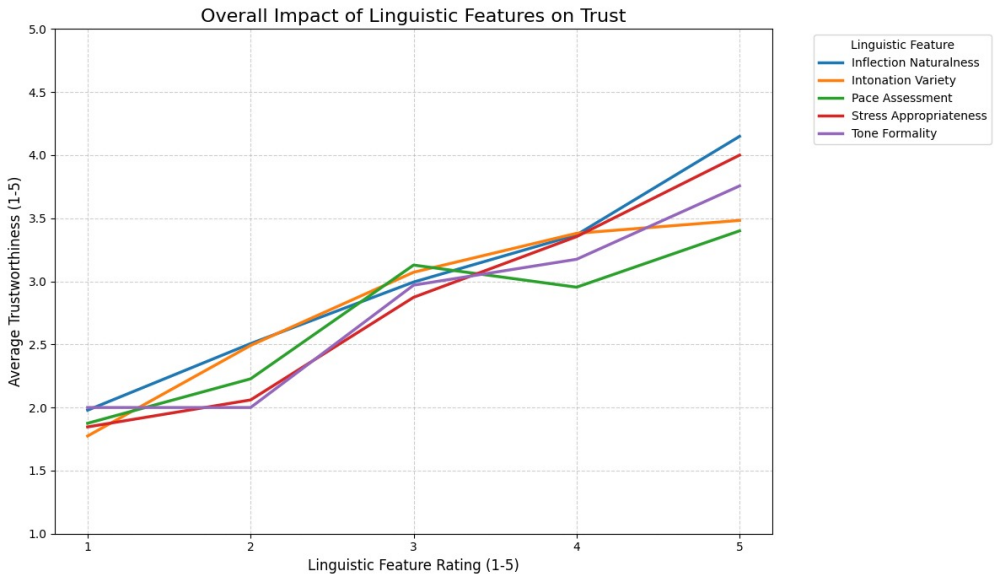

Figure 4: Impact of prosodic features on perceived trustworthiness.

Participants' trust ratings were strongly correlated with prosodic quality. As illustrated above, higher inflection and smoother intonation contours were associated with increased perceived trustworthiness, with a consistent upward trend, regardless of whether the clip was actually human or AI. In particular, the graph below demonstrates a clear positive relationship between inflection quality and credibility judgments.

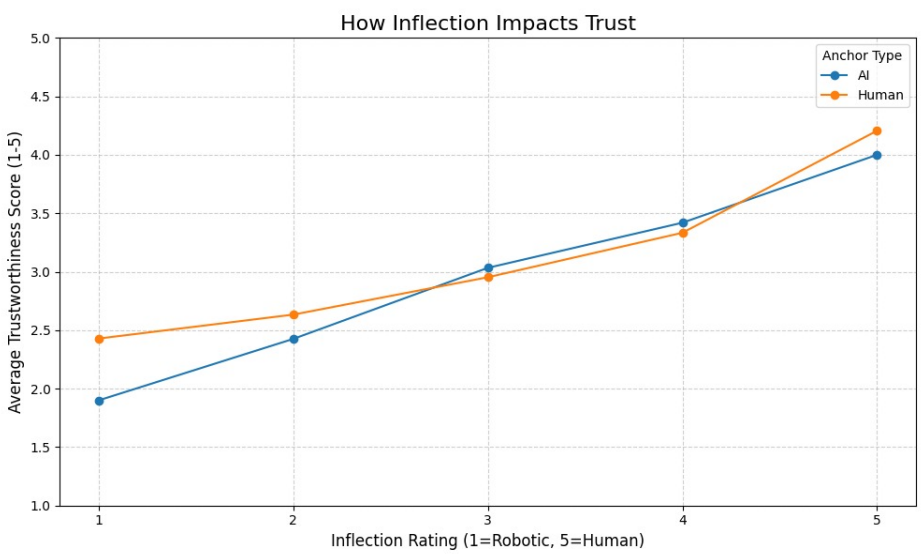

Figure 5: Relationship between inflection quality and perceived trustworthiness.

6.4 Impact of Speech Rate Manipulation

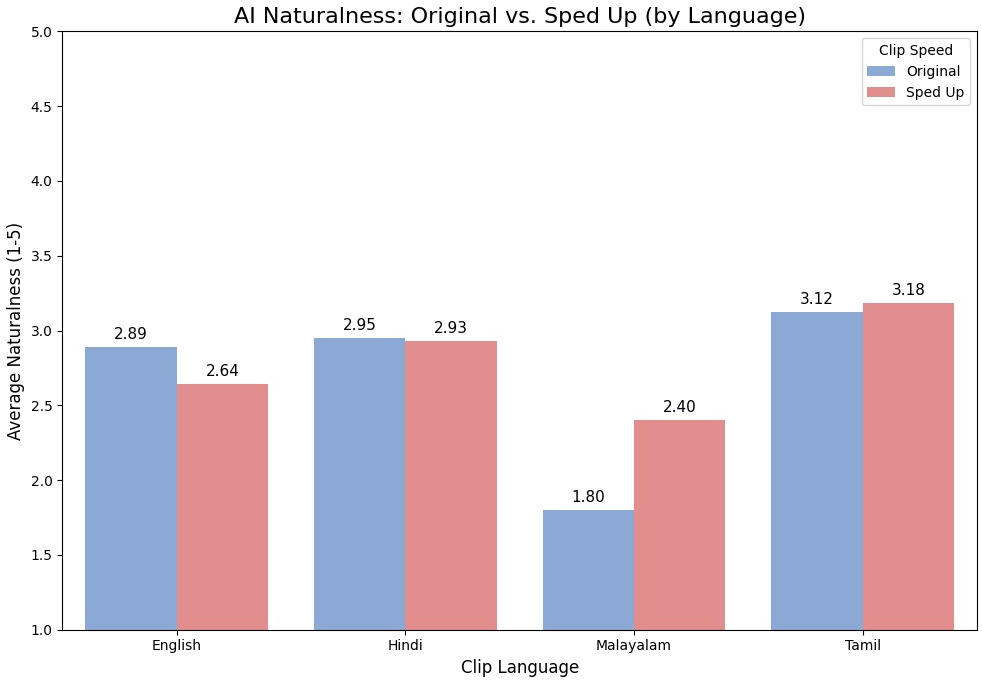

Figure 6: Effect of increasing AI speech rate on perceived naturalness.

Speeding up AI speech did not significantly improve perceived naturalness. While faster delivery made some clips sound closer to the rhythm of professional news speech, it did not resolve underlying issues of tonal flatness or unnatural stress placement. Listeners continued to identify sped–up AI clips as synthetic at similar rates.

6.5 Age-Based Variation

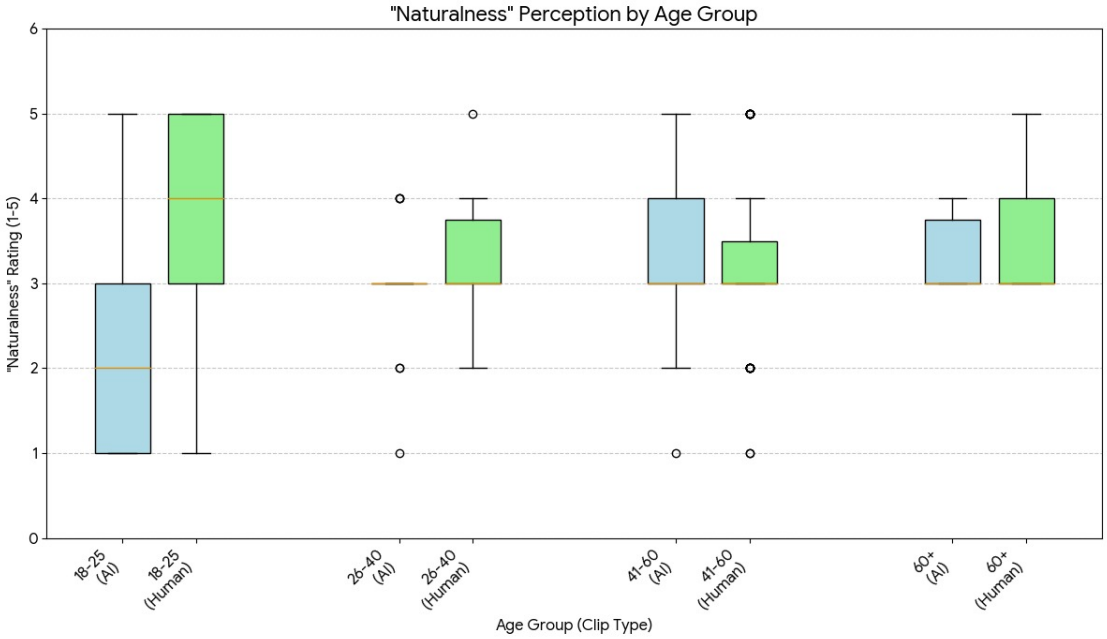

Figure 7: Naturalness perception by age group for Human vs. AI clips.

Age played a notable role in perceptual accuracy. Younger participants were more successful at identifying AI-generated speech. They rated AI clips as less natural on average. Older participants, by contrast, exhibited more variability and occasionally judged AI speech as natural, suggesting differences in familiarity with AI technologies or expectations of news delivery.

6.6 Effect of Mother Tongue Proficiency

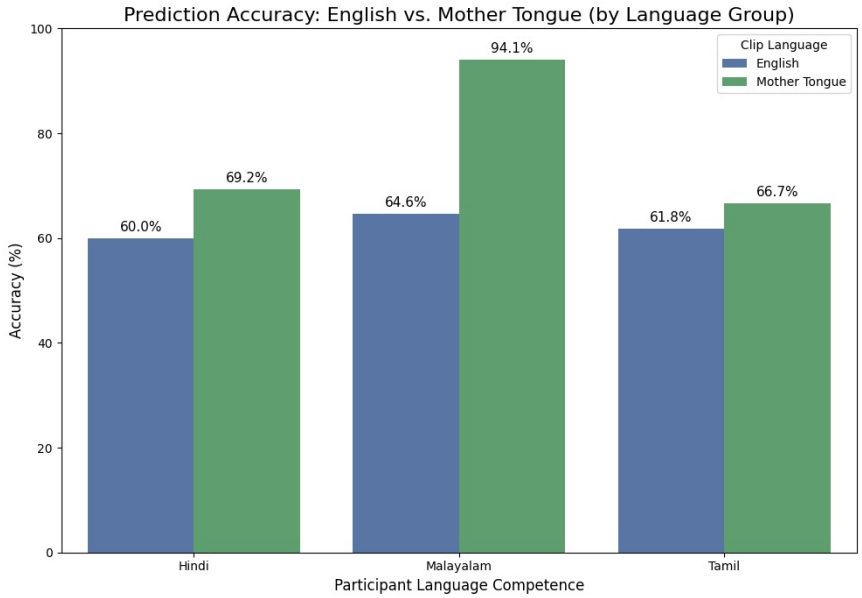

Figure 8: Prediction accuracy: English vs. mother tongue clips.

Language background influenced both accuracy and confidence. Participants performed substantially better when evaluating clips in their mother tongue. Familiarity with native prosodic patterns allowed them to detect subtle deviations that non-native listeners might overlook. Additionally, speakers from different linguistic groups prioritised different perceptual cues, indicating that prosodic salience varies across speech communities.

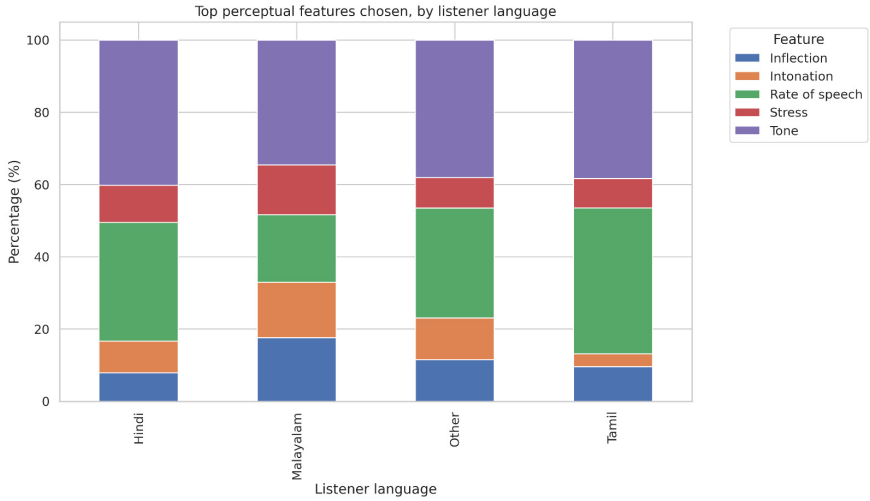

Figure 9: Perceptual cue preferences across language backgrounds.

6.7 Overall Patterns

In summary, human newscaster speech was consistently perceived as more natural, credible, and linguistically well-formed. AI speech, although intelligible and stable, lacked the prosodic richness required to convey authenticity. Listener judgments were strongly conditioned by prosodic cues, linguistic identity, and previous exposure to synthetic voices. These patterns collectively emphasise the centrality of prosody in distinguishing human and AI news delivery.

7 Statistical Analysis

The quantitative analysis focused on three aspects: (i) differences in mean ratings between human and AI clips, (ii) relationships between prosodic evaluations and perceived trustworthiness, and (iii) the impact of listener demographics and language background on accuracy and cue selection. Unless otherwise stated, statistical significance was evaluated at $\alpha = 0.05$.

7.1 Human–AI Differences in Mean Ratings

Human speech received higher average scores than AI speech on every feature, including naturalness, emotional impact, and trustworthiness. The magnitude of these differences was typically between 0.3 and 0.6 points on a 5-point scale, suggesting a consistent, moderate advantage for human newscasters in perceived prosodic quality and credibility.

7.2 Correlations Between Prosody and Trustworthiness

To quantify how strongly prosodic judgments influenced credibility, we computed both Pearson product–moment correlations and Spearman rank correlations between selected feature ratings. Key results are summarised below.

Naturalness and Trustworthiness. Overall perceived naturalness was strongly associated with trust judgments. Naturalness ratings were positively correlated with trustworthiness (Pearson $r = 0.65$, $p < 0.001$; Spearman $\rho = 0.64$, $p < 0.001$), indicating that more natural-sounding clips tended to be judged as more trustworthy. Naturalness alone accounts for approximately $r^2 \approx 0.42$ (42%) of the variance in trust scores.

Inflection and Naturalness. Inflectional naturalness was even more tightly linked to overall naturalness (Pearson $r = 0.76$, $p < 0.001$; Spearman $\rho = 0.74$, $p < 0.001$), explaining about 57% of its variance. This supports the pattern visible in Figure **??**, where higher inflection ratings correspond to higher trust scores for both human and AI anchors.

Rate of Speech and Urgency Perception. Perceived pace of speech showed a moderate positive correlation with urgency perception (Pearson $r = 0.47$, $p < 0.001$; Spearman $\rho = 0.42$, $p < 0.001$), suggesting that faster speech is reliably interpreted as more urgent (Figure **??**). This relationship, however, does not by itself guarantee higher trust.

Intonation Variety and Identification Accuracy. A small but statistically significant negative correlation was found between intonation variety ratings and the proportion of correctly identified AI clips (Pearson $r = -0.13$, $p \approx 0.008$; Spearman $\rho = -0.12$, $p \approx 0.016$). AI clips that were rated as having richer intonation were therefore slightly more likely to be mistaken for human speech.

7.3 Feature Importance for Detecting Artificiality

For correctly identified AI clips, listeners assigned higher importance scores to inflection and intonation, whereas misclassified AI clips showed lower average ranks for these features.

7.4 Effects of Speech Rate Manipulation

Changes in speed produced only small shifts (typically below 0.3 points) and in inconsistent directions across languages: English and Hindi clips became slightly less natural when sped up, while Malayalam and Tamil clips showed modest improvements. Taken together with the correlation results above, this suggests that altering rate of speech alone does not substantially improve perceived naturalness of AI news speech.

7.5 Age and Perception

Age-wise distributions of naturalness ratings point to systematic demographic differences. Across age groups, human clips received higher median naturalness scores than AI clips. Younger listeners (18–25) showed the largest separation between human and AI ratings and a wider spread for AI clips, consistent with greater sensitivity to synthetic artefacts. Older groups (41–60 and 60+) displayed narrower interquartile ranges and occasionally rated AI clips as comparably natural to human ones, suggesting more tolerant or less discriminating perceptions of synthesized speech. These trends are descriptive; due to sample size constraints, formal inferential testing across age bands was not conducted.

7.6 Language Background and Cue Selection

Prediction accuracy by language group demonstrates that participants were more accurate when evaluating clips in their mother tongue than in English. For example, Malayalam speakers achieved over 90% accuracy on Malayalam clips compared to mid-60% on English clips, while Hindi and Tamil speakers showed smaller but similar advantages for their respective languages. This supports the hypothesis that familiarity with native prosodic patterns facilitates detection of synthetic speech.

Tone was the most frequently selected perceptual feature across all groups, but Malayalam listeners relied more heavily on intonation and inflection, whereas Hindi and Tamil listeners showed a stronger emphasis on rate of speech. These patterns suggest that the relative salience of prosodic dimensions is shaped by listeners' linguistic backgrounds, and that strategies for identifying AI speech are not entirely universal.

8 Error Analysis

To understand why participants misclassified certain clips, we examined the subset of responses in which (i) AI clips were judged to be human and (ii) human clips were judged to be AI. These errors illuminate which prosodic cues listeners rely on and which artefacts in the speech signal lead to perceptual confusion.

8.1 Misclassification of AI Clips as Human

AI clips that were incorrectly labelled as human exhibited higher ratings for intonation variety and inflectional naturalness. As seen in Figure **??**, these misclassified clips scored lower on the "artificiality" ranking for features such as stress and tone. In particular, AI voices with smoother pitch contours or more consistent stress placement appeared more "human-like" to listeners, making them harder to detect. This is consistent with the negative correlation observed between intonation variety and identification accuracy: richer intonation tended to mask synthetic qualities.

Several AI voices also mimicked the formal register and neutral delivery style characteristic of news broadcasting. Because listeners already expect news anchors to maintain steady tone and controlled emotion, AI-generated clips that adhered closely to these conventions were more likely to be mistaken for genuine human speech.

8.2 Misclassification of Human Clips as AI

Human clips that were incorrectly judged as AI typically suffered from audio artefacts or prosodic deviations. These included inconsistent microphone quality, reduced pitch range, and atypically uniform stress patterns. In some cases, human anchors with naturally flatter intonation or a more clipped speaking style were perceived as synthetic, demonstrating that listeners sometimes mistake legitimate human variability for AI "roboticness."

Moreover, when content familiarity was lower (e.g., listeners hearing a non-mother-tongue clip), participants were more inclined to attribute unfamiliar prosodic patterns to artificial generation. This language-mismatch effect aligns with the higher accuracy rates observed for mother-tongue clips.

Overall, classification errors reflect a combination of prosodic cues, expectations about news speech, and differences in linguistic background rather than random guessing. These patterns highlight the nuanced perceptual boundaries between human and synthetic news delivery.

9 Sociolinguistic Interpretation

The findings of this study carry implications for how listeners relate linguistic cues to authenticity, trust, and identity in mediated speech. Because news broadcasting is a culturally loaded domain associated with authority, neutrality, and professional credibility, the perception of human versus AI voices is shaped not only by acoustic features but also by social expectations.

9.1 Prosody as a Marker of Authenticity

Prosodic features—especially tone, inflection, and intonation—function as indexical cues that listeners associate with human intentionality and emotional presence. Human anchors employ subtle shifts in pitch, timing, and emphasis that index involvement, seriousness, or empathy. AI voices, despite improvements in clarity, often lack these micro-variations, leading to lower perceived authenticity. The strong correlation between naturalness and trustworthiness reflects the close sociolinguistic linkage between sounding "human" and sounding "credible."

9.2 The Role of Linguistic Identity

Participants demonstrated higher accuracy and confidence when evaluating clips in their mother tongue. This suggests that prosodic norms are deeply internalised and culturally specific. Listeners draw on tacit knowledge of rhythm, pitch movement, and stress patterns characteristic of their speech community, enabling them to notice subtle unnatural cues in AI-generated speech. Differences across listener groups in cue selection (Figure **??**) further illustrate that linguistic identity shapes which features are considered perceptually salient.

9.3 Enregisterment of Newsreader Style

News broadcasting is an enregistered style: a recognisable and culturally codified way of speaking associated with institutional authority and media professionalism. This style involves a characteristic rhythmic pattern, strategic pitch contours, and a formal but warm tone. Human anchors acquire this style through training, socialisation, and exposure to genre norms. AI voices, in contrast, attempt to approximate this style algorithmically, often capturing the broad outline but missing the fine-grained prosodic cues that constitute the style's social meaning. The inability of AI anchors to fully perform this enregistered form contributes to the perceptual gap between human and synthetic news speech.

9.4 Expectations of News Register

News broadcasting is a codified register with its own prosodic conventions: measured pacing, formal tone, and controlled intonation. AI anchors that approximate these conventions blur the human–AI boundary, explaining why certain clips were misclassified. Conversely, human anchors whose delivery deviates from the expected "news voice" risk being perceived as artificial. This demonstrates how genre expectations interact with technological perception.

9.5 Stance, Intonation, and Perceived Credibility

Listeners interpret prosodic features as markers of stance—whether the speaker appears confident, neutral, engaged, or detached. Intonation patterns that rise, fall, or hold steady play a central role in signalling epistemic stance (confidence, certainty) and interpersonal stance (warmth, involvement). Human anchors use these cues to project authority and credibility, shaping how audiences evaluate the news. AI voices, by contrast, often produce flattened

or overly regular intonation contours, which signal detachment or lack of communicative intent. This mismatch in stance signalling helps explain why participants rated AI clips as less trustworthy even when the linguistic content was identical.

9.6 Trust, Familiarity, and Technological Attitudes

Younger participants showed greater accuracy and more pronounced distinctions between AI and human naturalness. This may reflect higher exposure to text-to-speech systems, greater metalinguistic awareness of synthetic prosody, or scepticism toward AI-generated media. Older participants' narrower rating ranges could stem from differing expectations of what constitutes "natural" news speech or less familiarity with AI artefacts.

Finally, the consistent finding that improved prosody leads to higher trust—and that AI voices with more natural-sounding inflection are perceived as more credible—illustrates that trust in mediated speech is not merely content-dependent. It is also shaped by listeners' beliefs about agency, intention, and human presence in the speech signal.

Overall, the sociolinguistic evidence suggests that AI-generated news voices face challenges not only in acoustic imitation but also in meeting culturally embedded expectations of human communication.

10 Conclusion

The findings of this study demonstrate that prosody is a critical determinant of how listeners perceive the naturalness, credibility, and authenticity of news speech. Across languages and demographic groups, human newscaster clips consistently received higher ratings for tone, inflection, stress, and intonation, and were perceived as significantly more trustworthy. Statistical analysis further revealed strong positive correlations between prosodic naturalness and trustworthiness, indicating that listeners interpret prosodic cues as signals of communicative intention and reliability.

While some AI voices approximated professional news register closely enough to be misclassified as human, the majority of AI clips displayed identifiable shortcomings: limited intonation range, mechanical stress placement, and reduced expressive variation. These deficiencies resulted in lower naturalness and trust scores, and contributed to higher rates of correct identification of synthetic speech. Manipulating speech rate alone had minimal effect, suggesting that surface-level prosodic adjustments cannot compensate for deeper limitations in expressive modelling.

Importantly, perceptual accuracy varied across linguistic groups and age groups. Participants performed substantially better in their mother tongue, underscoring the role of culturally embedded prosodic expectations in detecting artificial speech. Younger listeners also showed greater sensitivity to synthetic artefacts, highlighting generational differences in familiarity with and expectations of AI-generated media.

Overall, the study suggests that although AI anchors can deliver intelligible and stylistically consistent speech, their prosodic limitations impede their ability to fully replicate the sociolinguistic nuances of human news delivery. The human anchor remains, at present, not

merely a conduit of information but a culturally situated voice that signals trust, identity, and authenticity—dimensions that AI systems have yet to convincingly emulate.

11 Implications for Media Practice

The findings of this study raise important considerations for how AI-generated voices should be deployed in news broadcasting. While AI anchors offer advantages in speed, scalability, and operational efficiency, they currently fall short in conveying the prosodic richness and social meaning that listeners associate with credible news delivery. This has direct implications for media institutions seeking to integrate AI voices into their workflows.

First, the persistent gap in perceived naturalness and trustworthiness suggests that AI anchors may not be suitable for high-stakes or sensitive contexts, such as political reporting, crisis communication, or public health announcements. These domains rely heavily on affective and interpersonal cues—subtle shifts in tone, emphasis, and pacing—that signal empathy, seriousness, and human judgement. AI systems that fail to reproduce such cues risk diminishing the perceived legitimacy of the content.

Second, AI anchors may be more appropriate for low-stakes, informational, or repetitive announcements where emotional nuance is less critical. Examples include weather updates, sports summaries, stock tickers, or automated service bulletins. In these contexts, listeners may prioritise clarity and consistency over prosodic expressiveness.

Finally, organisations considering the use of AI anchors must weigh efficiency gains against the potential erosion of audience trust. Because trust is shaped not only by content accuracy but also by the indexical cues of human presence, overreliance on AI voices may weaken the relational dimension of news consumption. A hybrid model—where AI handles routine segments and human anchors deliver interpretive or sensitive content—may offer a more balanced path forward.

12 Present Shortcomings

Despite the robustness of the survey design, several methodological and empirical limitations constrain the generalisability of our findings:

- **Limited linguistic diversity of stimuli.** The dataset contained relatively few high-quality clips in Indic languages, especially Malayalam and Tamil. This restricted our ability to analyse language-specific prosodic effects across a balanced set of stimuli.

- **Inconsistency in AI audio quality.** The AI anchor clips varied substantially in synthesis quality across different voice models. These uncontrolled differences may have inflated or reduced detection accuracy for certain voices.

- **Small and uneven participant demographics.** Although 80 respondents participated, the age distribution was uneven and certain linguistic groups were underrepresented. This reduces the statistical power of cross-group comparisons.

- **Absence of acoustic pre-processing.** No noise filtering or loudness normalisation was applied to the audio stimuli. Background noise or differing microphone quality may have inadvertently influenced perceived naturalness and trust.
- **Incomplete topic normalisation.** While politically charged or emotionally loaded content was avoided, differences in topic structure and information density across clips were not strictly controlled and may have introduced cognitive bias.
- **No direct acoustic analysis.** The study relied solely on perceptual judgments without measuring acoustic parameters such as pitch range, pause structure, or spectral variation, limiting insight into the phonetic sources of perceived artificiality.
- **Limited number of synthetic models.** Only a small subset of contemporary AI anchors were included. As synthesis quality varies widely across systems, the results cannot be generalised to all existing technologies.

13 Future Scope

Future research can extend and refine this study in several directions:

- **Expansion of the multilingual dataset.** Incorporating a larger and more balanced set of human and AI clips across multiple Indian languages will enable richer cross-linguistic analysis of prosodic cues.
- **Larger and demographically diverse participant pool.** Gathering responses from participants across different regions, age groups, and linguistic communities will improve the generalisability of perceptual trends and strengthen statistical inference.
- **Acoustic and phonetic analysis.** Extracting quantitative measures of pitch range, intonation contours, stress placement, and temporal structure would allow linking of perceptual judgments with specific acoustic features.
- **Modelling credibility using joint linguistic–perceptual features.** Integrating acoustic features with survey-derived judgments could facilitate predictive modelling of trustworthiness in human and AI voices.
- **Longitudinal or exposure-based studies.** Examining how repeated exposure to AI-generated news affects listeners' expectations, skepticism, and trust could reveal how perceptions evolve over time.
- **Genre and register variation.** Extending the study beyond news—e.g., to interviews, podcasts, or narrative reading—would illuminate how register-specific prosodic expectations influence perceptions of artificiality.
- **Discourse-level analysis.** Investigating information structuring, phrasing, and rhetorical emphasis in AI versus human speech may reveal deeper linguistic differences not captured by segment-level prosody.